

Fig. 2: An actual-size, single-side PCB for the industrial timer circuit

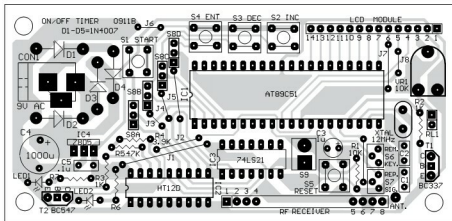


Fig. 3: Component layout for the PCB in Fig. 2

first process starts second process and so on), one can use a number of such monostable blocks to make a sequential timer. But these circuits do not include additional features like digital display, system failure indication, remote operation and alarms. Since the precision and accuracy of these timer circuits depend upon the value of the resistor-capacitor components that may deviate, we might not get the exact and precise timing. To enhance the programmable timer for generating precise timing and additional features, microcontrollers (embedded controllers) are used with peripheral devices.

Some of the features of programmable industrial on/off timer presented here include:

1. Time set from 1 to 60 seconds (can be extended)
2. 'On' time and 'off' time can be programmed (from 1 to 60 seconds)

3. Repeat (continuous) and single operation
4. Fully remote-controlled within 100-metre range
5. User-friendly front-panel controls and display panel with LCD
6. Emergency stop buttons (on control panel as well as on remote)
7. Provision of potential-free relay contacts for connecting any 230VAC at 10A or 28V DC at 10A device/application

Circuit description

The complete hardware circuit is divided into two sections—programmable timer main circuit and remote control transmitter and receiver units. The main circuit is a fully-functional standalone unit with front-panel control switch and LCD display. Remote control transmitter and receiver are add-on units. The receiver outputs are

connected with respective control input pins of main circuit. Receiver output facilitates user to program and control timer operation using remote within 100-metre range.

Main circuit. Fig. 1 shows the circuit diagram of industrial on/off timer. The circuit comprises microcontroller AT89C51 (IC1), LCD, transistors and a few discrete components. The data pins D0 through D7 of LCD are connected with port pins P2.0 through P2.7 of microcontroller AT89C51. The control pins register select (RS), read/write (R/W) and enable (E) are connected with port P3 pins P3.0, P3.1 and P3.2, respectively. Preset VR1 is connected with pin 3 of LCD for contrast control. Switches S1 through S4 are connected to port pins P1.0 through P1.3 by 4PDT toggle switch S8. Two single-pole double-throw (SPDT) switches S6 and S7 are connected at P3.5 and P3.6. Switch S6 selects for either remote or keypad control, and switch S7 selects control for either repeat or single mode operation. Port pin P3.7 drives relay through transistors T1. Diode D5 acts as free-wheeling diode for relay RL1.

Power-on reset is provided by the combination of resistor R1 and capacitor C3. Switch S5 is used for manual reset or emergency stop function. A 12MHz quartz crystal along with two 33pF capacitors provides clock pulse to the microcontroller AT89C51.

The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C4 and regulated by IC 7805 (IC4). Capacitor C5 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R7 limits the current through LED1.

An actual-size, single-side PCB for the programmable industrial on/off timer is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it minimises assembly time and errors. Carefully assemble the